

Time-changed Lévy setup

$$\begin{aligned}\log S_t &= \log S_0 + (r - q)t + Y_t \\ Y_t &= X_{\tau(t)} + \omega t \\ S_t &= S_0 \exp\{(r - q)t + X_{\tau(t)} + \omega t\}\end{aligned}$$

Characteristic exponent

$$\begin{aligned}\phi_X(u; t) &= \mathbb{E}[e^{iuX_t}] = \exp\{t\kappa_X(u)\} \\ \kappa_X(u) &= ibu - \frac{1}{2}\sigma^2 u^2 + \int_{\mathbb{R} \setminus \{0\}} (e^{iux} - 1 - iux \mathbf{1}_{\{|x| < 1\}}) \nu(dx) \\ \Psi_X(u) &= -\kappa_X(u)\end{aligned}$$

CGMY Lévy density

$$\begin{aligned}\nu_{\text{CGMY}}(dx) &= C \frac{e^{-Mx}}{x^{1+Y}} \mathbf{1}_{\{x>0\}} dx + C \frac{e^{-G|x|}}{|x|^{1+Y}} \mathbf{1}_{\{x<0\}} dx \\ C &> 0, \quad G > 0, \quad M > 0, \quad Y < 2\end{aligned}$$

CGMY characteristic function

$$\begin{aligned}\kappa_{\text{CGMY}}(u) &= C\Gamma(-Y) [(M - iu)^Y - M^Y + (G + iu)^Y - G^Y] \\ \phi_{\text{CGMY}}(u; T) &= \exp\{T\kappa_{\text{CGMY}}(u)\}\end{aligned}$$

NIG characteristic exponent

$$\begin{aligned}\kappa_{\text{NIG}}(u) &= i\mu u + \delta \left(\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + iu)^2} \right) \\ \alpha &> |\beta|, \quad \delta > 0 \\ \phi_{\text{NIG}}(u; T) &= \exp\{T\kappa_{\text{NIG}}(u)\}\end{aligned}$$

Gamma subordinator

$$\begin{aligned}\Gamma_T &\sim \text{Gamma}\left(\frac{T}{\nu}, \nu\right) \\ \mathbb{E}[\Gamma_T] &= T, \quad \text{Var}(\Gamma_T) = \nu T \\ \mathbb{E}[e^{-\lambda \Gamma_T}] &= (1 + \nu \lambda)^{-T/\nu} \\ \mathbb{E}[e^{z \Gamma_T}] &= (1 - \nu z)^{-T/\nu}\end{aligned}$$

Independent Gamma time change

$$\begin{aligned}\phi_{X_{\Gamma_T}}(u) &= \mathbb{E}[e^{iuX_{\Gamma_T}}] \\ \phi_{X_{\Gamma_T}}(u) &= \mathbb{E}[\mathbb{E}[e^{iuX_{\Gamma_T}} \mid \Gamma_T]] \\ \phi_{X_{\Gamma_T}}(u) &= \mathbb{E}[e^{\Gamma_T \kappa_X(u)}] = (1 - \nu \kappa_X(u))^{-T/\nu} \\ \kappa_{X \circ \Gamma}(u) &= -\frac{1}{\nu} \log(1 - \nu \kappa_X(u)) \\ \phi_{X_{\Gamma_T}}(u) &= \exp\{T \kappa_{X \circ \Gamma}(u)\}\end{aligned}$$

NIG plus Gamma

$$\begin{aligned}\kappa_{\text{NIG} \circ \Gamma}(u) &= -\frac{1}{\nu} \log(1 - \nu \kappa_{\text{NIG}}(u)) \\ \phi_{\text{NIG} \circ \Gamma}(u; T) &= (1 - \nu \kappa_{\text{NIG}}(u))^{-T/\nu} \\ \phi_{\text{NIG} \circ \Gamma}(u; T) &= \exp\{T \kappa_{\text{NIG} \circ \Gamma}(u)\}\end{aligned}$$

CGMY plus Gamma

$$\begin{aligned}\kappa_{\text{CGMY} \circ \Gamma}(u) &= -\frac{1}{\nu} \log(1 - \nu \kappa_{\text{CGMY}}(u)) \\ \phi_{\text{CGMY} \circ \Gamma}(u; T) &= (1 - \nu \kappa_{\text{CGMY}}(u))^{-T/\nu} \\ \phi_{\text{CGMY} \circ \Gamma}(u; T) &= \exp\{T \kappa_{\text{CGMY} \circ \Gamma}(u)\}\end{aligned}$$

Effective characteristic exponent

$$\begin{aligned}\kappa_{\text{eff}}(u) &= \kappa_X(u) \\ \kappa_{\text{eff}}(u) &= -\frac{1}{\nu} \log(1 - \nu \kappa_X(u)) \\ X &\in \{\text{NIG}, \text{CGMY}\}\end{aligned}$$

Martingale correction

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[e^{-rt} S_t] &= S_0 e^{-qt} \\ \mathbb{E}^{\mathbb{Q}}[e^{X_t^{\text{eff}} + \omega t}] &= 1 \\ \omega t &= -\log \mathbb{E}^{\mathbb{Q}}[e^{X_t^{\text{eff}}}] \\ \omega &= -\kappa_{\text{eff}}(-i)\end{aligned}$$

Log-price characteristic function

$$\begin{aligned}\log S_T &= \log S_0 + (r - q + \omega)T + X_T^{\text{eff}} \\ \phi_{\log S_T}(u) &= \mathbb{E}^{\mathbb{Q}}[e^{iu \log S_T}] \\ \phi_{\log S_T}(u) &= \exp(iu [\log S_0 + (r - q + \omega)T] + T\kappa_{\text{eff}}(u))\end{aligned}$$

Carr–Madan damped call transform

$$\begin{aligned}k &= \log K \\ c_T(k) &= e^{\alpha k} C_T(k) \\ \widehat{C}_\alpha(v, T) &= \int_{-\infty}^{\infty} e^{ivk} c_T(k) dk \\ \widehat{C}_\alpha(v, T) &= \frac{e^{-rT} \phi_{\log S_T}(v - i(\alpha + 1))}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v} \\ C_T(k) &= \frac{e^{-\alpha k}}{\pi} \int_0^{\infty} \text{Re}\left[e^{-ivk} \widehat{C}_\alpha(v, T)\right] dv\end{aligned}$$

FFT grid

$$\begin{aligned}v_j &= j\eta, \quad k_m = -b + m\lambda, \quad \lambda\eta = \frac{2\pi}{N} \\ C_T(k_m) &\approx \frac{e^{-\alpha k_m}}{\pi} \sum_{j=0}^{N-1} \text{Re}\left[e^{-iv_j k_m} \widehat{C}_\alpha(v_j, T)\right] \eta w_j\end{aligned}$$